

SAMASTHA - IELTS MOCK

An AI Powered Preparation Platform

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This project implements a comprehensive IELTS examination system that assesses English proficiency through **Listening, Reading, Writing, and Speaking** modules.

It simulates the official IELTS environment, offering realistic practice with timed tests, instant feedback, and performance tracking.

Designed for non-native speakers, it supports academic, immigration, and professional needs. The platform emphasizes accessibility, scalability, and a user-friendly interface for diverse users.

Introduction

This IELTS application offers both individual practice tests for each module — **Listening, Reading, Writing, and Speaking** — as well as full-length tests covering all sections.

It provides detailed feedback for each test, maintains test history, and tracks user performance through interactive graphs.

The system is designed to help users improve effectively with targeted practice and comprehensive performance insights.

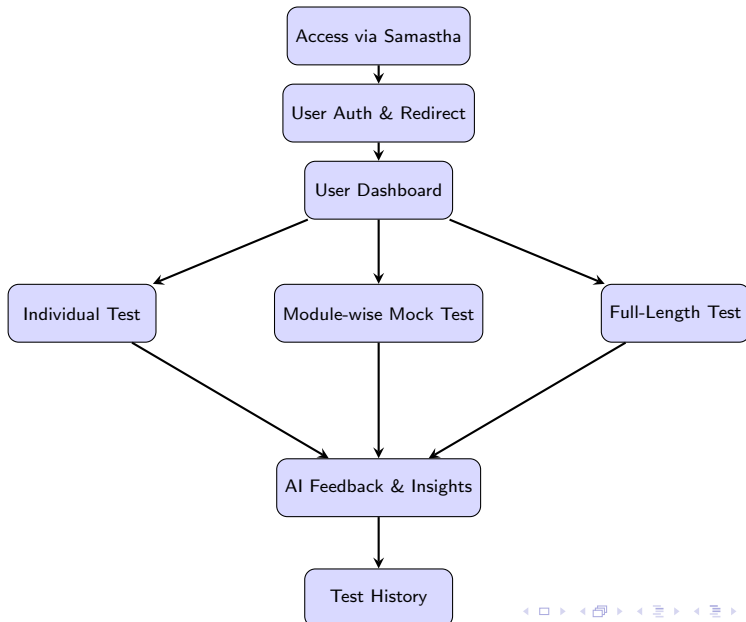
Existing Platforms:

- **British Council IELTS Prep:** Offers practice tests but lacks real-time feedback and candidate-specific tracking.
- **IELTS Liz / Magoosh:** Focus on content and tips but do not offer interactive test-taking environments or progress analytics.
- **Cambridge IELTS Books:** Provide quality practice material, but are static and not personalized.

Our Differentiation:

- Designed to train and monitor the progress of candidates who enroll through our service.
- Empowers customers (consultants/institutions) to track performance, provide guided improvement, and ensure candidates are exam-ready.
- Acts as a bridge between candidates' dreams and actionable, measurable training.

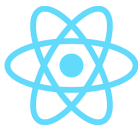
Proposed Method - Flowchart



Proposed Method

- **Samastha Access:** Users must be registered via the Samastha partner platform. Only verified users can access the IELTS portal.
- **User Dashboard:** Central hub for accessing different test types, viewing performance data, and navigating features.
- **Test Types:**
 - **Individual Test:** Practice a single part from any module (e.g., a single Listening section).
 - **Module-wise Mock Test:** Full mock test of one complete module (e.g., all 3 Reading parts).
 - **Full-Length Test:** Simulates the official IELTS exam with all four modules in sequence.
- **AI Feedback:** After each test, the system generates detailed, module-specific feedback using AI — covering accuracy, grammar, and suggestions for improvement.
- **Test History:** All test attempts are saved with scores and insights, helping users track progress and spot trends over time.

Technologies Used



React.js



Tailwind CSS



Python Flask



MySQL



OpenAI API & Claude



Google Speech-to-Text



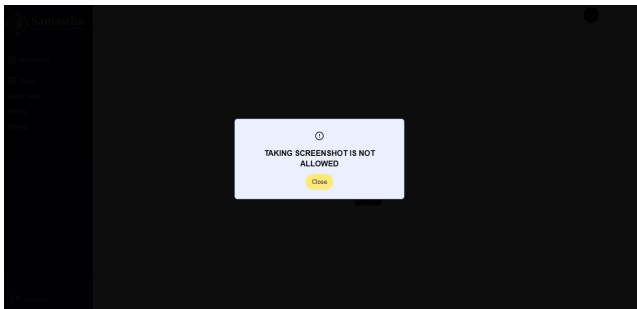
JWT Auth



Vercel (Deployment)

Result

- Developed UI using **React JS** + **Tailwind CSS** for all modules.
- Implemented routing and timer controls for individual and full-length tests.
- Designed instruction, question, and result pages with responsive layouts.



Screenshot of Screenshot Disabling Function

Result

```
// Function to check if current page is a feedback page
const isFeedbackPage = () => {
  const path = location.pathname

  // Use regex patterns to match feedback routes including those with dynamic parameters
  const feedbackPatterns = [
    /^\/fulltest\/feedback(\/.*)?$/,
    /^\/mock\/writing\/feedback(\/.*)?$/,
    /^\/mock\/listening\/feedback(\/.*)?$/,
    /^\/mock\/reading\/feedback(\/.*)?$/,
    /^\/practice\/writing\/feedback\/.*$/,
    /^\/practice\/listening\/feedback1\/.*$/,
    /^\/practice\/listening\/feedback2\/.*$/,
    /^\/practice\/listening\/feedback3\/.*$/,
    /^\/practice\/listening\/feedback4\/.*$/,
    /^\/practice\/reading\/feedback\/.*$/,
    /^\/speaking_feedback(\/.*)?$/,
    /^\/fulltestspeaking_feedback$/,
    /^\/speakingprepfeedback$/,
  ]

  // Return true if any pattern matches the current path
  const isFeedback = feedbackPatterns.some((pattern) => pattern.test(path))
  return isFeedback
}

// Handle screenshot prevention
const handleScreenshotPrevention = (event) => {
  // Skip prevention if on a feedback page
  if (isFeedbackPage()) {
    console.log("Feedback page detected, skipping screenshot prevention.")
    return
  }

  if (
    event.key === "PrintScreen" || // PrtSc key
    (event.ctrlKey && event.key === "PrintScreen") || // Ctrl + PrtSc
    (event.altKey && event.key === "PrintScreen") || // Alt + PrtSc
    (event.shiftKey && event.metaKey) || // Windows + Shift (Meta for Mac)
    (event.key === "S" && event.shiftKey && event.metaKey) || // Windows + Shift + S (Mac: Command)
    (event.key === "S" && event.shiftKey && event.ctrlKey) // Ctrl + Shift + S
  ) {
    event.preventDefault()
    setIsScreenShot(true)
    setCardData("TAKING SCREENSHOT IS NOT ALLOWED")
    setCard(true)
  }
}

// Handle right-click prevention
const handleRightClickPrevention = (event) => {
```

Result

```
// Handle right-click prevention
const handleRightClickPrevention = (event) => {
  // Skip prevention if on a feedback page
  if (isFeedbackPage()) return

  event.preventDefault()
  console.log("Right-click disabled!")
  setIsScreenshot(true)
  setCardData("RIGHT CLICK IS DISABLED")
  setCard(true)
}

// Handle dev tools prevention
const handleDevToolsPrevention = (event) => {
  // Skip prevention if on a feedback page
  if (isFeedbackPage()) return

  if (
    event.key === "F12" ||
    (event.ctrlKey && event.shiftKey && event.key === "I") || // Ctrl+Shift+I
    (event.ctrlKey && event.shiftKey && event.key === "C") // Ctrl+Shift+C
  ) {
    event.preventDefault()
    console.log("DevTools blocked!")
    setIsScreenshot(true)
    setCardData("FUNCTIONAL KEYS ARE BLOCKED")
    setCard(true)
  }
}

// Set up event listeners with useEffect
useEffect(() => {
  // Add event listeners
  document.addEventListener("keydown", handleScreenshotPrevention)
  document.addEventListener("contextmenu", handleRightClickPrevention)
  document.addEventListener("keydown", handleDevToolsPrevention)

  // Clean up event listeners when component unmounts
  return () => {
    document.removeEventListener("keydown", handleScreenshotPrevention)
    document.removeEventListener("contextmenu", handleRightClickPrevention)
    document.removeEventListener("keydown", handleDevToolsPrevention)
  }
}, [location.pathname])
```



The screenshot displays the 'Test History' page of the Samastha IELTS Mock application. On the left is a dark blue sidebar with the Samastha logo and navigation links: Dashboard, Study, Demo Video, History (highlighted), and Settings. At the bottom of the sidebar is a 'log out' link. The main content area has a light gray header with a user profile icon labeled 'TS' and a filter dropdown set to 'All Tests'. Below the header, a horizontal bar contains filters for 'All (9)', 'Full Test (1)', 'Listening (2)', 'Speaking (1)', 'Reading (2)', and 'Writing (2)'. The test history is listed as follows:

Test Type	Date	Score	Action
Reading Test	Apr 13, 03:07PM	Score: 0 Band	>
Reading Test	Apr 14, 11:23AM	Score: 2.5 Band	>
Writing Test	Apr 11, 12:10PM	Score: 6.5 Band	>
Reading Test	Apr 10, 03:10PM	Score: 3.5 Band	>
Reading Test	Apr 10, 03:10PM	Score: 3.5 Band	>
Full IELTS Test	Apr 10, 03:04PM	Overall Score: 1 Band 23.5 18.5 17.5	View Feedback >
Writing Test	Apr 10, 03:03PM	Score: 0 Band	>
Writing Test		Score: 0 Band	>

Screenshot of Test History Page

- Dashboard
- Study
 - Lecture Videos
 - Practice Test
- History
- Help
- Settings
- Log out

Back

Practice Tests

All (12)

Listening (4)

Speaking (3)

Reading (3)

Writing (2)

All

Short Answer Questions

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

Start Test →

Part-1

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Identifying Info & Claims

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Task-1_Comparison Data

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Short Answer Questions

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Part-1

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Identifying Info & Claims

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Task-1_Comparison Data

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

Short Answer Questions

Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Attempts: 0/10

→

screenshot of Practice tests cards

The screenshot displays the 'Lecture videos' section of the SIRIAM OVERSEAS interface. On the left is a dark blue sidebar with navigation links: Dashboard, Study (with a sub-link for Lecture Videos), History, Help, Settings, and Log out. The main content area has a light purple header with the title 'Lecture videos' and a user profile icon. Below the title is a filter bar showing 'All (32)' selected, with other filters for Listening (6), Speaking (9), Reading (10), and Writing (9). An 'All Videos' dropdown is also present. The video thumbnails are arranged in a 2x3 grid. Each thumbnail features the SIRIAM robot character, the text 'IELTS Listening Short Answer', and a duration of 05:29. The first row includes 'Short Answer Questions', 'Part-1', and 'Identifying info & Claims'. The second row includes 'Short Answer Questions', 'Sentence completion', and 'Multiple choice'. Each video card has a brief description of the listening task.

SIRIAM OVERSEAS

Dashboard ▾

Study ▴

Lecture Videos

Practice Test

History

Help

Settings

Log out

Lecture videos

All (32) Listening (6) Speaking (9) Reading (10) Writing (9) All Videos ▾

IELTS Listening
Short Answer

05:29

Short Answer Questions
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

IELTS Listening
Short Answer

03:29

Part-1
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

IELTS Listening
Short Answer

05:29

Identifying info & Claims
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

IELTS Listening
Short Answer

05:29

Short Answer Questions
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

IELTS Listening
Short Answer

03:29

Sentence completion
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

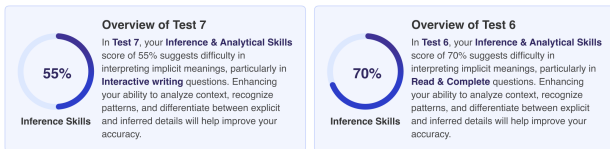
IELTS Listening
Short Answer

05:29

Multiple choice
Focuses on the ability to listen for facts such as places, prices or times, heard in the recording.

Screenshot of Lecture Videos Page

Comparison of Test 7 and Test 6



Question Types	Total Questions	Test 7 Performance		Test 6 Performance	
Read & Select	15-17	✗ 2/17	✓ 15/17	✗ 0/15	✓ 15/15
Fill in the Blanks	6-8	✗ 1/6	✓ 5/6	✗ 2/8	✓ 6/8
Read & Complete	3-6	✗ 2/6	✓ 4/6	✗ 4/6	✓ 2/6
Interactive Reading	15-18	✗ 9/15	✓ 6/15	✗ 4/18	✓ 14/18

Conclusion

In Test 7, your Inference & Analytical Skills score was lower due to incorrect answers in **Interactive Reading (9/15)** and **Read & Select (2/17)** questions. These areas significantly impacted your performance. However, in Test 6, you showed improvement in these question types, answering **14/18 Interactive Reading** and **15/15 Read & Select** correctly. While **Fill in the Blanks (1/6)** remained a challenge in both tests, your overall accuracy was higher in Test 6, with fewer mistakes, leading to a better inference skill score.

Screenshot of Compare Page

Conclusion

- Developed a complete, AI-powered IELTS preparation platform that simulates real exam conditions across all four modules.
- Empowered users with dynamic practice environments, instant feedback, and progress tracking dashboards.
- Successfully integrated modern technologies such as **React.js**, **Tailwind CSS**, **Python FastAPI/Flask**, and **AI/NLP services**.
- Achieved a seamless, scalable solution for both learners and institutions to enhance English proficiency in a structured, data-driven way.
- Future improvements could include mobile app development, peer evaluations, adaptive testing, and LMS integration.

1 Figma UI/UX Design:

<https://www.figma.com/design/9f2Ea52IGTKrhAiZyGWPkn/TEST-PREP-AI-TOOLS?node-id=4943-29809&t=lyBbDLxd5luwzPoH-0>

2 OpenAI API Documentation:

<https://platform.openai.com/docs>

3 Google Cloud Speech-to-Text API:

<https://cloud.google.com/speech-to-text/docs>

4 React.js Official Documentation:

<https://reactjs.org/docs/getting-started.html>

5 Tailwind CSS Documentation:

<https://tailwindcss.com/docs>

6 MySQL Official Documentation:

<https://dev.mysql.com/doc/>

7 Project Contributions and Documentation: B Mallesh (Self Documentation, Architecture Planning, Final Presentation)

"The beautiful thing about learning is that no one can take it away from you."

— B.B. King

Experience the Platform Live:

<https://www.samasthabroad.com>

Thank You!